Low Error Is Not a Safe Alarm

Certified Miss-Rate Guarantees for Machine-Failure Alerts, Tested on Simulated Jet Engines and 343,944 Real Hard Drives

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ABSTRACT

Machine-learning models that predict the remaining useful life (RUL) of machines are usually judged by their average error. A maintenance team, however, acts on alarms, and the dangerous error is a machine that fails before the next inspection because no alarm was raised. We first reproduced a public benchmark on NASA's C-MAPSS turbofan simulation (Random Forest MAE 13.39 cycles vs. 13.38 reported). We then measured alarms instead of averages and introduced the **guard**, a conformal alert rule that sets its threshold from a small set of recorded failures so that the **miss rate among failing machines is at most a chosen α**, with a finite-sample guarantee. In a pre-registered study (8 hypotheses, frozen with SHA-256 hashes, Holm-corrected bootstrap tests), the guard targeting α = 10% missed 7.9% [4.9, 11.3] of failing engines on the official C-MAPSS test sets, with 10.0 and 5.6 percentage points fewer false alarms than split-conformal and conformalized-quantile-regression bounds. When a model was moved to a different fleet, the guarantee broke (27.1% missed), but recalibrating the threshold on 20 failures from the new fleet restored it (10.0% [7.1, 13.3]). On a real fleet of 343,944 hard drives with 5,730 failures, evaluated three months after calibration, the industry SMART rule missed 25.6% of failures, while the guard missed 7.4% [6.0, 9.0]. It stayed valid on manufacturers never seen in training once it was recalibrated on 20 of their failures (8.3%). The guarantee is honest about cost, though. Because many drives fail without any SMART warning, a 90% catch guarantee required alerting 48% of healthy drives, while a 5% inspection budget certified a catch rate of about 74% (76% achieved). Seven of eight hypotheses were supported. The exception was that a plain point-prediction alarm on the same fleet missed 12.9% [8.5, 17.7], not significantly more than 10%.

Keywords: predictive maintenance · remaining useful life · conformal prediction · distribution shift · NASA C-MAPSS · Backblaze · pre-registration

1. Introduction

Unexpected machine failures stop production lines, ground aircraft, and destroy data. Predictive maintenance tries to replace fixed service schedules with decisions based on sensor data: if a model can tell that a machine is close to failure, it can be serviced just in time, neither too early (wasting useful life) nor too late (an in-service failure) [4]. The standard research benchmark for this task is NASA's C-MAPSS turbofan simulation [1, 2], where a model predicts the *remaining useful life* (RUL) of an engine, in operating cycles, from its sensor history.

Papers on C-MAPSS almost always report the average size of the error, as mean absolute error (MAE) or root-mean-squared error (RMSE). This project started the same way. Our first report compared Linear Regression, Random Forest and XGBoost on the FD001 sub-dataset and quoted a public benchmark in which Random Forest reached an MAE of 13.38 cycles [3]. We first **reproduced that benchmark on our own server** (Section 3), obtaining an MAE of 13.39 cycles. But reproducing it raised the question this paper is about:

> **Does a low average error make a safe maintenance alarm?** A maintenance team does not act on an average. At each inspection it decides, machine by machine, whether to service now or wait until the next inspection. The error that matters is a machine that fails before the next inspection because the model said it was fine. We call this a *miss*.

We show that the answer is no. A model with a competitive RMSE, alerting when its predicted RUL falls below the inspection interval, misses a large share of the engines that are about to fail, and far more when the fleet changes (new operating conditions or a new fault mode). We then build and test an alert rule, the **guard**, that turns any RUL or risk model into an alarm with a **certified miss rate**. The user chooses the acceptable miss rate α (for example 10%), and a short calculation on a small set of past failures sets the alert threshold so that, under a stated assumption, at most a fraction α of failing machines are missed. The method comes from conformal prediction [5–8]. Our contribution is to apply it to the right quantity (misses among machines that are about to fail, not average coverage), to measure what it costs in false alarms, to show when the guarantee breaks, and to show how many real failures are needed to repair it.

Because the original report's main limitation was that C-MAPSS is simulated, we also test everything on a **real fleet**: daily SMART sensor records from 343,944 hard drives operated by Backblaze in 2024–2025 [9], including 5,730 real failures. All hypotheses, settings and analysis code were pre-registered (frozen with SHA-256 hashes) before the test data were evaluated.

Contributions. (1) An independent reproduction of the public FD001 benchmark. (2) Evidence that RUL error metrics hide unsafe alarms, in-fleet and across fleets. (3) The guard: a distribution-free alert rule with a finite-sample miss-rate guarantee, plus a minimum-data rule and a prediction of how much the realised miss rate varies. (4) A budget mode that tells an operator what catch rate a fixed inspection budget certifies. (5) A pre-registered evaluation on 4 simulated fleets, 12 cross-fleet transfers, and a real drive fleet tested three months into the future and on unseen manufacturers.

2. Background

2.1 NASA C-MAPSS

C-MAPSS simulates turbofan engines that start with unknown wear and degrade until failure [1]. It has four sub-datasets (Table 1). Each row is one engine at one cycle: three operating settings and 21 sensor readings. Training engines run until failure, so their RUL at every cycle is known. Test engines stop at a random point before failure, and the true RUL at that point is provided separately. FD002 and FD004 mix six operating conditions; FD003 and FD004 contain two fault modes (high-pressure compressor and fan degradation).

Table 1. C-MAPSS sub-datasets [1, 2]. "At risk" = official test engines whose true RUL is ≤ 30 cycles.

Fleet	Conditions	Fault modes	Train engines	Test engines	At risk (test)
FD001	1	1	100	100	25
FD002	6	1	260	259	61
FD003	1	2	100	100	20
FD004	6	2	249	248	53

2.2 Backblaze hard-drive data

Backblaze publishes one record per drive per day for every drive in its data centres: model, capacity, a failure flag, and SMART (Self-Monitoring, Analysis and Reporting Technology) attributes such as reallocated sectors (SMART 5) or uncorrectable errors (SMART 187) [9]. A drive

"fails" when it is removed because it stopped working or was about to. We used January 2024 to March 2025 (456 days, about 125 million drive-days). The industry rule of thumb used by Backblaze itself flags a drive when any of SMART 5, 187, 188, 197 or 198 is above zero [10]; we use it as a real-world baseline.

2.3 Conformal prediction

Conformal prediction is a family of methods that wrap any model and give guarantees that hold for finite samples and need no assumption about the data distribution other than *exchangeability* (the calibration examples and the new example are like random draws from the same population, in any order) [5, 6]. The usual version gives a prediction interval that contains the truth with probability at least $1 - \alpha$, averaged over all examples. Variants that hold separately within each class are called label-conditional or Mondrian [6, 7]. Conformalized quantile regression (CQR) [8] is a strong recent variant that adapts the interval width to each input. Conformal prediction intervals for RUL on C-MAPSS were introduced in [11]. Robinson [12] made them asymmetric, spending more of the error budget on over-estimates because late maintenance is the costlier mistake; this is the closest work to ours. Both guarantee coverage averaged over all prediction points. What we add is: (i) the guarantee is placed directly on the decision that matters, the misses among units that are about to fail; (ii) its false-alarm cost is measured against those interval methods; (iii) it is tested across fleets and on a real fleet, with a rule for how many failures are needed to recalibrate.

3. Reproducing the public benchmark

The original report quoted MAE and RMSE values for three models on FD001 from a public repository [3] and correctly warned that they were not the team's results. We re-ran the experiment from the official NASA files. Following common practice we capped the training target at 125 cycles (an engine early in life shows no measurable wear, so its exact RUL cannot be predicted), dropped seven sensors that are constant in FD001, and scored the prediction at the last cycle of each of the 100 official test engines against the true RUL.

Table 2. FD001 official test set: public benchmark [3] versus our reproduction (`run_fd001.py`). Without the 125-cycle cap the errors roughly double (MAE 23.6–25.5), a detail the original report did not mention.

Model	MAE [3]	MAE (ours)	RMSE [3]	RMSE (ours)	R^2 (ours)
Linear Regression	17.64	17.62	22.01	21.90	0.722
Random Forest	13.38	13.39	18.13	18.28	0.806
XGBoost	13.43	13.17	18.46	18.17	0.809

The reproduction matches to within 0.3 cycles, so the reported numbers are credible. However, in the same Random Forest run, 57 of the 100 test predictions were too *late* (predicted RUL above the true RUL), which is the dangerous direction. The rest of this paper studies that direction directly.

4. Methods

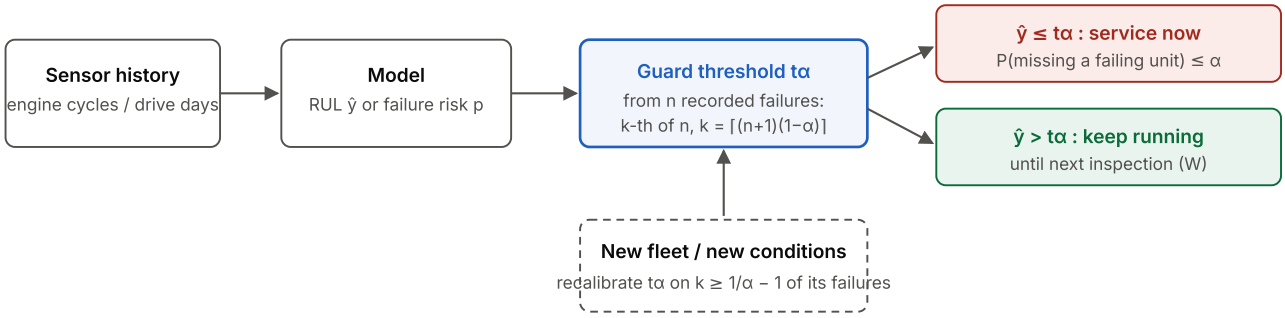


Figure 1. The guard. Any model that ranks units by how close they are to failure is turned into an alarm with a certified miss rate. The only extra input is a small set of past failures. When the fleet changes, the threshold (not the model) is recomputed from a few failures of the new fleet.

4.1 From RUL prediction to an alarm

We model a fleet that is inspected every W time units (cycles for engines, days for drives). At an inspection, a unit is **at risk** if its true RUL $\leq W$: if it is not serviced now, it will fail before the next inspection. An **alert rule** maps the model's output for a unit to "service now" or "wait". Two error rates matter:

- **Miss rate** = share of at-risk units that got no alert (in-service failures). This is the safety metric.
- **False-alarm rate** = share of units that are not at risk but got an alert (wasted inspections). This is the cost metric.

The primary setting is $W = 30$ and target miss rate $\alpha = 0.10$; $W \in \{20, 30, 50\}$ cycles (C-MAPSS) or $\{7, 30\}$ days (Backblaze), and $\alpha \in \{0.05, 0.10, 0.20\}$ are reported as secondary.

4.2 Models

C-MAPSS. Sensors are standardised within each operating condition (conditions found by k-means on the three settings), using statistics from the fleet's own data but never its failure labels, so the same step is possible for a newly deployed fleet. Features are the 14 informative sensors, their 10-cycle rolling mean and 20-cycle rolling slope, and the cycle count (43 features). The point model is XGBoost regression [13] on the RUL capped at 125. A second XGBoost model with the quantile loss gives a lower quantile for CQR.

Backblaze. For each drive and day we use 23 SMART raw values, plus the 7-day and 30-day change of the 13 error counters, plus capacity (50 features). The model is an XGBoost classifier for "fails within W days". The drive model name was deliberately left out, so the model can be applied to an unseen manufacturer. In development, adding it did not reduce the false-alarm cost.

4.3 Alert rules compared

Table 3. Alert rules. $\hat{y}$ = predicted RUL; for Backblaze, alerts trigger when the predicted failure probability is high, and the rules are mirrored.

Rule	Alert when	Origin
Point	$\hat{y} \le W$	The natural use of a RUL model
Margin	$\hat{y} \le W + \text{RMSE}_{\text{cal}}$	Engineering "safety margin"
Conformal lower bound	$\hat{y} - q \le W$, q = conformal quantile of $(\hat{y} - y)$	Standard split conformal, one-sided [5]
CQR lower bound	$\hat{y}_\alpha - e \le W$	Conformalized quantile regression [8]
Guard (ours)	**$\hat{y} \le t_\alpha$, threshold from at-risk calibration units**	**Label-conditional conformal [6, 7] on misses**
SMART rule (drives)	any of SMART 5/187/188/197/198 > 0	Industry practice [10]
Default / F1-optimal (drives)	p ≥ 0.5, or p ≥ threshold maximising F1 on calibration	Common ML practice

4.4 The guard and its guarantee

Take n calibration units that were observed while at risk (one snapshot per unit, drawn uniformly from its last W time units before failure) and compute the model's score for each: $s_1, \ldots, s_n$. Sort them and set

$$k = \lceil (n+1)(1-\alpha) \rceil, \quad t_\alpha = s_{(k)} \text{ (the } k\text{-th smallest)}, \quad \text{alert iff } \hat{y} \le t_\alpha.$$

> **Guarantee.** If a new at-risk unit is exchangeable with the n calibration units, then $\text{P(miss)} = \text{P}(s_{\text{new}} > t_\alpha) \le \alpha$. *Proof sketch:* by exchangeability the rank of s_{new} among all $n+1$ scores is uniform on $\{1, \ldots, n+1\}$; a miss requires rank $> k$, which has probability $(n+1-k)/(n+1) \le \alpha$. With no ties the miss rate is also at least $\alpha - 1/(n+1)$, so the guard is not wastefully cautious [5, 6].

Three practical consequences follow and are tested below:

1. **Minimum data.** If $k > n$ no threshold exists and the only safe rule is "always alert". A 90% guarantee needs at least $n = \lceil 1/\alpha \rceil - 1 = 9$ recorded failures, and a 95% guarantee needs 19.
2. **Spread.** The guarantee holds on average over calibration sets. For one particular calibration set, the realised miss rate follows a Beta$(n+1-k, k)$ distribution [14], so small calibration sets give noisy (sometimes worse than α) results even when the guarantee holds.
3. **Why not the usual conformal bound?** The standard lower bound guarantees coverage averaged over *all* units, most of which are healthy and far from failure. Nothing controls its error on the at-risk units specifically: it can be far too cautious (many false alarms) or not cautious enough. The guard spends its whole error budget on the units that matter.

Budget mode (drives). An operator may instead have a fixed inspection budget, for example 1% of healthy drives per month. The threshold is then the 99th percentile of calibration scores of healthy drives, and the guard's argument run in reverse gives a **certificate**: if m of the n calibration failures fall below the threshold, the expected miss rate is at most $(m+1)/(n+1)$.

Recalibration under fleet change. Exchangeability fails when the new fleet differs from the calibration fleet. The fix we test is simple: keep the model, but recompute t_α from k failures recorded in the new fleet ($k \in \{5, 10, 20, 40\}$ engines; $\{10, 20, 40, 80\}$ drives).

4.5 Experimental design

Development (all choices of features, models and settings) used only the C-MAPSS *training* files, split by engine into fit / calibration / evaluation (50/25/25%, 30 repetitions), and only the first half of 2024 (P1) of the drive data. **Confirmatory tests** were run once, after freezing:

- **C-MAPSS, same fleet:** model fitted on 70% of a fleet's training engines and calibrated on the other 30%; evaluated on the official test engines (last observed cycle). Repeated 50 times with different splits.
- **C-MAPSS, cross-fleet:** the same model applied to each of the other three fleets' official test engines (12 transfers), with the source calibration, and recalibrated with k failures from the target fleet's training file.
- **Backblaze, future:** fitted on P1 (Jan–Jun 2024), calibrated on P2 (Jul–Sep 2024), evaluated on P3 (Oct–Dec 2024), with labels looking up to 30 days ahead into 2025. Five model seeds.
- **Backblaze, unseen manufacturer:** each of Seagate, HGST, Toshiba and WDC in turn is removed from fitting and calibration and evaluated on its P3 drives, with and without recalibration on k of its P2 failures.

The engine or drive is the independent unit everywhere: one snapshot per unit, no unit in two roles, and confidence intervals from a 10,000-draw bootstrap over units (stratified by fleet), with each unit's outcome averaged over repetitions. Eight hypotheses (Table 4) were fixed in `prereg.json` together with SHA-256 hashes of all code and data before any confirmatory run. A hypothesis is **supported** only if its 95% CI criterion is met *and* its one-sided bootstrap p-value survives Holm correction across all eight. Non-inferiority claims use a margin of 0.05 (the miss rate's upper 95% bound must be ≤ 0.15 when $\alpha = 0.10$).

Disclosure. Before freezing, the official FD001 test set was used for the benchmark reproduction (Section 3), and the FD001 and FD003 test sets were used once in a pilot with a different model and features. No settings were tuned on them afterwards. The software is Python 3.11, scikit-learn 1.9.1, XGBoost 3.2.0, pandas 3.0.6 and DuckDB 1.5.6, run on a 2-core ARM cloud server. Code: [repository URL].

5. Results

Table 4. Pre-registered hypotheses (primary setting $W = 30$, $\alpha = 0.10$, margin 0.05). Estimates are pooled miss rates (H3: difference in false-alarm rate, guard minus comparator) with 95% bootstrap CIs over engines/drives; p-values are one-sided and Holm-corrected across all eight. Computed by the frozen `analyze.py` (SHA-256 verified).

ID	Claim	Estimate	95% CI	p (Holm)	Result
H1	C-MAPSS in-fleet: point-alert miss rate > alpha	0.129	[0.085, 0.177]	0.105	Not supported
H2	C-MAPSS in-fleet: guard miss rate <= alpha + margin	0.079	[0.049, 0.113]	<0.001	Supported
H3a	C-MAPSS in-fleet: guard false-alarm rate < conformal_lb	-0.100	[-0.121, -0.080]	<0.001	Supported
H3b	C-MAPSS in-fleet: guard false-alarm rate < cqr_lb	-0.056	[-0.071, -0.042]	<0.001	Supported
H4	C-MAPSS cross-fleet: source-calibrated guard miss > alpha	0.271	[0.226, 0.316]	<0.001	Supported
H5	C-MAPSS cross-fleet: guard recalibrated on 20 target failures, miss <= alpha + margin	0.100	[0.071, 0.133]	0.004	Supported
H6	Backblaze, 3 months later: SMART-rule miss rate > alpha	0.256	[0.230, 0.283]	<0.001	Supported
H7	Backblaze, 3 months later: guard miss rate <= alpha + margin	0.074	[0.060, 0.090]	<0.001	Supported
H8	Backblaze, unseen manufacturer: guard recalibrated on 20 failures, miss <= alpha + margin	0.083	[0.071, 0.097]	<0.001	Supported

5.1 Same fleet: low error, but which alarm?

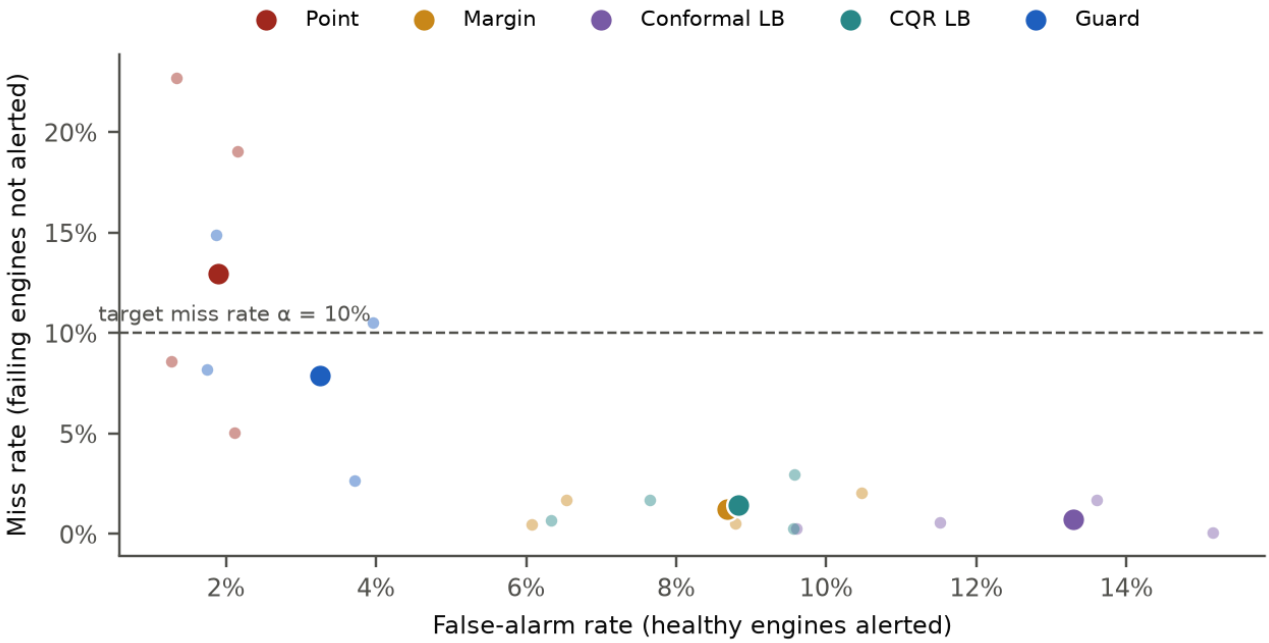


Figure 2. C-MAPSS, same fleet, official test engines (W = 30, α = 10%). Large dots: pooled over the four fleets; small dots: each fleet. Only the guard sits at the requested miss rate; the interval methods buy unrequested extra safety with 3–4× more false alarms.

On the official test engines of the fleet they were trained on, all five rules are compared in Figure 2 and Table 5. The point rule, alerting when the predicted RUL is at most 30 cycles, missed 12.9% of the 159 at-risk engines (averaged over 50 calibration splits). That is more than the 10% target, but the CI [8.5, 17.7] includes 10%, so **H1 is not supported**: on its own fleet, a good RUL model is not dramatically unsafe. It is, however, unpredictable. It missed 22.7% in FD001 and 19.1% in FD004 but only 5.0% in FD002, and nothing in the RMSE tells the user which case they are in.

The guard missed 7.9% [4.9, 11.3] (**H2 supported**), and its miss rate followed the requested α at every setting (Table 6): 3.0%, 7.9% and 16.3% for α = 5%, 10% and 20% at W = 30. The two standard conformal bounds also had low miss rates, but only by being far more cautious than requested (0.7% and 1.4% missed when 10% was allowed), and they paid for it in false alarms: 13.3% and 8.8% of healthy engines alerted, versus 3.3% for the guard (**H3a and H3b supported**, differences −10.0 and −5.6 percentage points). In other words, the usual conformal intervals do not deliver the miss rate the user asked for; they deliver an unknown, smaller one at an unknown cost. The guard spends exactly the error budget it was given.

Table 5. C-MAPSS, same fleet, official test sets (W = 30, α = 0.10; 50 calibration splits). Miss = failing engines without alert; FA = healthy engines alerted.

Fleet	At risk	Point miss / FA	Margin	Conformal LB	CQR LB	Guard
FD001	25	22.7% / 1.3%	1.7% / 6.5%	0.6% / 11.5%	1.7% / 7.7%	14.9% / 1.9%
FD002	61	5.0% / 2.1%	0.5% / 8.8%	0.1% / 15.2%	0.3% / 9.6%	2.7% / 3.7%
FD003	20	8.6% / 1.3%	0.5% / 6.1%	0.3% / 9.6%	0.7% / 6.3%	8.2% / 1.8%
FD004	53	19.1% / 2.2%	2.1% / 10.5%	1.7% / 13.6%	3.0% / 9.6%	10.5% / 4.0%
Pooled	**159**	**12.9% / 1.9%**	**1.2% / 8.7%**	**0.7% / 13.3%**	**1.4% / 8.8%**	**7.9% / 3.3%**

Table 6. Miss rate of the guard versus the requested α (C-MAPSS same fleet, pooled). The guard tracks α; the other rules do not respond to it.

W (cycles)	α = 5%	α = 10%	α = 20%	Conformal LB at α = 10%	Point
20	4.1%	8.9%	17.1%	0.2%	13.0%
30	3.0%	7.9%	16.3%	0.7%	12.9%
50	1.1%	3.9%	11.6%	1.2%	8.7%

5.2 A new fleet breaks every rule, and 20 failures fix the guard

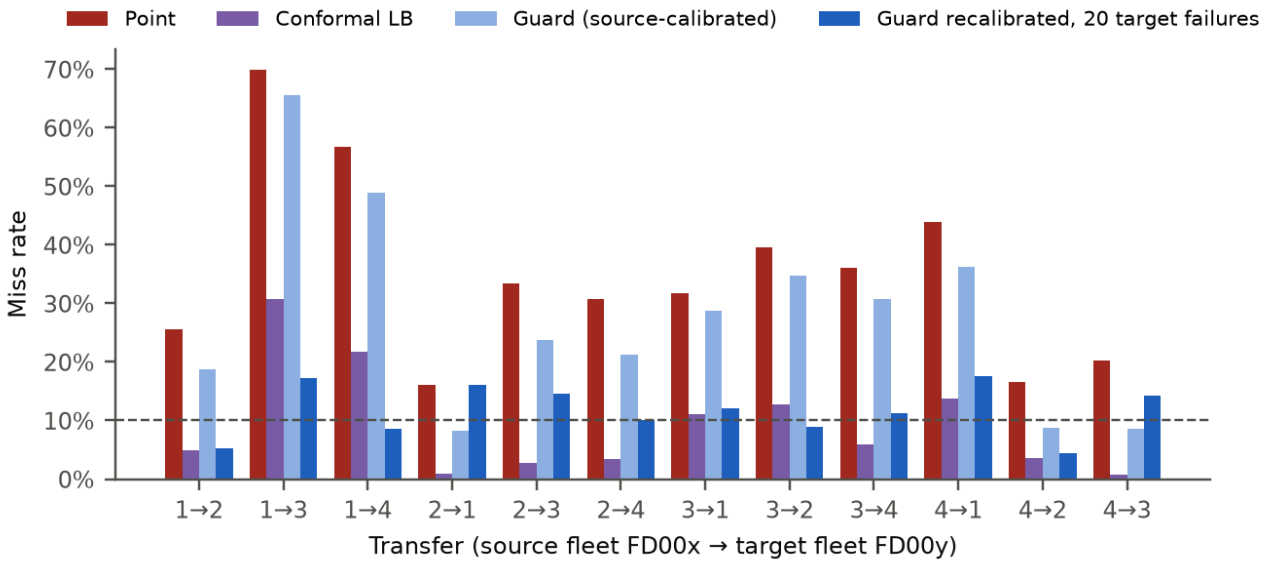


Figure 3. Miss rate when a model trained and calibrated on one fleet is used on another (official test engines, α = 10%). Recalibrating the guard's threshold with 20 failures from the target fleet (dark blue) brings the pooled miss rate back to 10.0%; individual transfers range from 4% to 18%, the spread expected from only 20 calibration failures.

Applying a model to one of the other three fleets (12 transfers, Figure 3) is where alarms fail. The point rule missed 34.2% of failing engines and up to 70% (FD001 → FD003, a fault mode never seen in training). The guard calibrated on the old fleet missed 27.1% [22.6, 31.6] (**H4 supported**): its guarantee assumes the new failures look like the old ones, and they do not. The conformal lower bound, which is very cautious, still missed up to 31% on individual transfers.

Recalibrating only the threshold, with the model unchanged, on 20 failures recorded in the new fleet brought the pooled miss rate to 10.0% [7.1, 13.3] (**H5 supported**). Individual transfers ranged from 4% to 18%, which matches the spread predicted by the Beta distribution (Section 4.4). Figure 4 shows the data rule in action. With 5 failures no 90% threshold exists, so the guard alerts every engine (100% false alarms). With 10 failures it already reaches 9.5% misses, and more failures mainly reduce false alarms (10.2% → 8.4% → 7.5% for 10, 20, 40 failures). For α = 5%, 10 failures are not enough, again as the theory says.

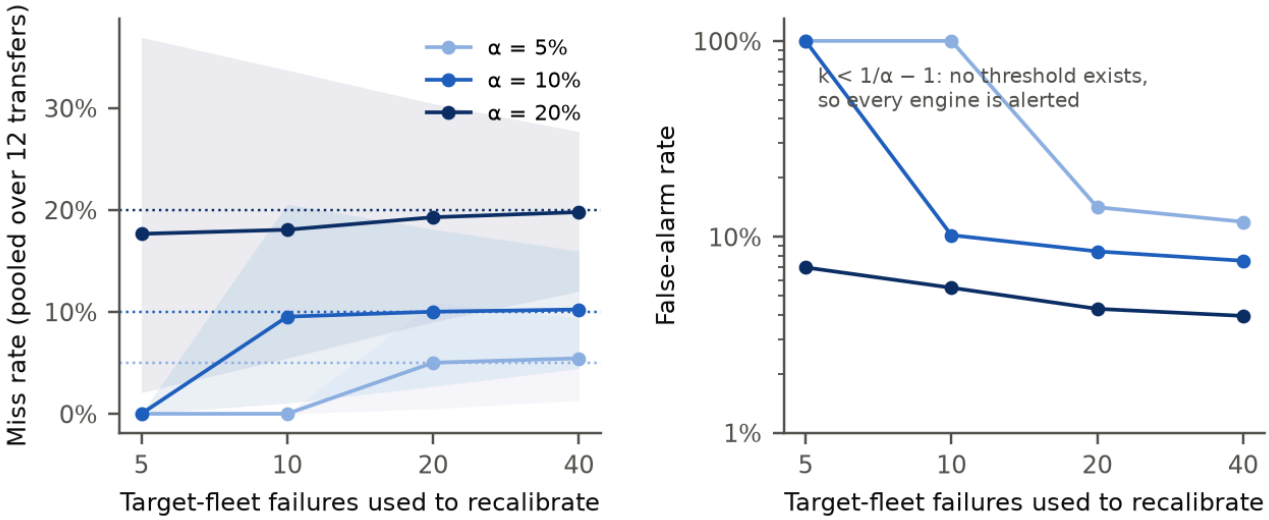


Figure 4. Recalibration with k target-fleet failures, pooled over the 12 transfers. Left: miss rate, with the 10–90% range predicted by the Beta(n+1−k, k) distribution for a single calibration set (shaded). Right: false alarms. Below $\lceil 1/\alpha \rceil - 1$ failures no threshold exists and every engine is alerted.

5.3 A real fleet: 343,944 hard drives

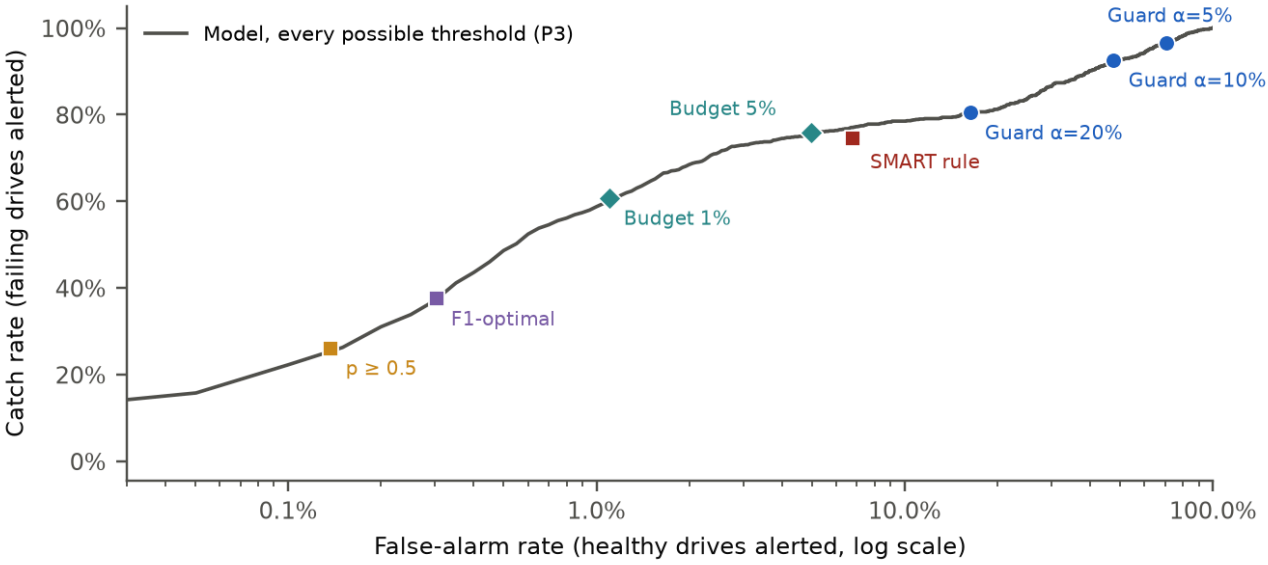


Figure 5. Backblaze, October–December 2024 (model fitted on January–June, thresholds set on July–September). The curve shows every threshold of one model; markers show the rules of Table 7. The steep upper part is the price of catching failures that SMART does not reveal.

Table 7 shows the drive results three months into the future. The industry SMART rule caught 74.4% of failing drives at a 6.8% false-alarm rate, so it missed 25.6% [23.0, 28.3] (**H6 supported**). A model used the usual ML way was worse for safety: a 0.5 probability threshold missed 73.9%, and the F1-optimal threshold 62.5%, because failures are rare and these thresholds are tuned for accuracy, not for catching failures. The guard calibrated on July–September 2024 and applied to October–December 2024 missed 7.4% [6.0, 9.0] when 10% was allowed (**H7 supported**), and 3.4% and 19.6% for α = 5% and 20%. So the guarantee survived three months of real fleet change.

The cost, however, is large and the guard makes it visible. Figure 5 shows the whole trade-off. About a quarter of failing drives show essentially nothing in SMART before they fail (Backblaze found the same in 2016 [10]), so catching 90% of failures meant alerting 47.6% of healthy drives in a

month, and 80% meant 16.3%. For a realistic budget the operator should use budget mode: at 1% and 5% inspection budgets the model caught 60.7% and 75.8% of failing drives. The calibration certificates promised at most 42.8% and 26.4% misses, and the realised misses were 39.3% and 24.2%. At a 5% budget the model therefore matches the SMART rule's catch rate with fewer inspections, and it comes with a certificate that the SMART rule does not have.

Table 7. Backblaze, fit Jan–Jun 2024, calibrated Jul–Sep 2024, evaluated Oct–Dec 2024 (5,140 at-risk snapshots over 5 seeds; about 300,000 healthy drives). "Cert." = miss-rate bound promised by the calibration data before seeing the test period.

Rule	Miss (W = 30 d)	False alarms	Cert.	Miss (W = 7 d)	False alarms
SMART rule (industry)	25.6%	6.8%		18.2%	6.9%
Model, p ≥ 0.5	73.9%	0.1%		61.5%	0.1%
Model, F1-optimal threshold	62.5%	0.3%		65.2%	0.1%
Budget mode, 1% inspected	39.3%	1.1%	≤ 42.8%	30.5%	1.0%
Budget mode, 5% inspected	24.2%	5.0%	≤ 26.4%	16.9%	5.1%
Guard, α = 20%	19.6%	16.3%		17.1%	4.6%
Guard, α = 10%	**7.5%**	**47.6%**		**9.6%**	**30.0%**
Guard, α = 5%	3.4%	70.6%		4.4%	58.6%

Unseen manufacturers. When a manufacturer was removed from fitting and calibration, the shift went the other way than in C-MAPSS. The unfamiliar drives looked risky to the model, and the source-calibrated guard alerted almost all of them (1.0% misses, 97.5% false alarms). Recalibrating on 20 failures of the new manufacturer kept the guarantee (8.3% [7.1, 9.7], **H8 supported**) and lowered false alarms to 75.6%. That is still very high, because a model trained without, for example, Seagate (the largest maker) is simply a poor model for Seagate drives. The guard can certify a model's alarms but cannot make a weak model strong. Per-manufacturer results also show the limit of a fleet-wide guarantee. With the full model, the 5%-budget certificate (26.4%) was met for HGST (13.4%) and Seagate (20.8%) but not for Toshiba (33.2%) or WDC (40.7%), whose failures are less visible in SMART.

6. Discussion

What a judge of predictive-maintenance models should ask. Our reproduction confirms that tree models reach MAE ≈ 13 cycles on FD001. But the same models, used as alarms, missed between 5% and 23% of failing engines depending on the fleet, and between 16% and 70% after a fleet change, with no warning from the error metrics. RMSE and MAE average over all predictions, most of which concern healthy machines far from failure. The decision that matters concerns the few machines near failure.

Why the guard works, and when it does not. The guard puts the whole guarantee on the units that matter, so it delivers the requested miss rate and leaves the rest of the error budget unused instead of wasting it on false alarms. That is why it needed 2.7–4 times fewer false alarms than the interval methods at the same or better safety level. Its guarantee rests on one assumption, that new failures resemble the calibration failures. The cross-fleet experiments show this is not a technicality: it is the main way predictive-maintenance alarms fail in practice. The practical rule is simple and cheap. Record the first $\lceil 1/\alpha \rceil - 1$ failures in any new fleet (9 for a 90% guarantee, about 20 for a stable one) and recompute one number.

Honest costs. On real drives the guard turns a vague model into an explicit, certified trade-off between safety and inspection cost. The trade-off was steep because SMART data are silent before

many failures. This is a property of the sensors, not of the method, and it is useful to know: no threshold on these data can catch 90% of failures cheaply, and a guaranteed rule says so instead of hiding it in an F1 score.

Answering the original report's questions. (1) Useful sensors: for engines, the slowly drifting temperatures and pressures (rolling means and slopes); for drives, reallocated/pending sectors and their recent change. (2) Lowest error: XGBoost ≈ Random Forest < Linear Regression (Table 2). (3) How often predictions are too late: 57 of 100 FD001 test engines had over-estimated RUL. With a point alarm that means 5–23% missed failures within a fleet and up to 70% across fleets. (4) Generalisation beyond simulation: yes for the guarantee (real drives, future period, unseen manufacturers, after recalibration), but the cost of the guarantee depends on how much the sensors reveal.

7. Limitations

- **Exchangeability is an assumption, not a fact.** The guarantee holds when new failing units resemble the calibration failures. Our cross-fleet and cross-manufacturer tests show exactly how far it breaks when they do not; in practice the threshold should be recalibrated whenever the fleet, its operating conditions or its sensors change, and the realised miss rate should be monitored.
- **The guarantee is on average over the fleet.** It does not promise the same miss rate for every sub-group (for example, every drive model); our drive results show large differences between manufacturers. Calibrating separately per group (Mondrian calibration) fixes this but needs at least $1/\alpha - 1$ failures per group.
- **Official C-MAPSS test truncation.** The official test engines stop at points chosen by NASA, not uniformly within the last W cycles as in our calibration. This mismatch is realistic and part of what H2 tests, but it means the in-fleet test is slightly harder than the theory assumes.
- **Simulated versus real.** C-MAPSS is a simulation. The drive fleet is real, but hard drives are not jet engines, and SMART data are noisier and often silent before failure. Neither dataset includes maintenance actions, which in a real deployment would remove at-risk units from the data.
- **One model family.** All results use gradient-boosted trees. The guard works with any score, and better models move the whole trade-off curve, but we did not test deep sequence models.
- **Cost model.** We report misses and false alarms separately instead of combining them with assumed prices; the right α depends on the real cost of an in-service failure versus an inspection, which differs between industries.

8. Safety, ethics and research integrity

No equipment was damaged; all data are public (NASA open data [1, 2]; Backblaze drive stats, free to use with attribution [9]). The drive data contain serial numbers but no personal information. Numbers quoted from other work (Table 2, left columns) are labelled as such and were reproduced before being used. Hypotheses, settings and analysis code were frozen with SHA-256 hashes before the test data were evaluated, and every pre-registered result is reported, including those not supported. The pilot that touched two official test sets before freezing is disclosed in Section 4.5. The method supports, and does not replace, engineering judgement; it is not certified for aviation or other safety-critical use.

9. Conclusion

A low average error does not make a safe alarm. We reproduced a public C-MAPSS benchmark and then showed, in a pre-registered study on four simulated engine fleets and 343,944 real hard drives, that a simple conformal alert rule, the guard, delivers the miss rate it promises among failing machines. It uses far fewer false alarms than standard conformal intervals, it stays valid three months into the future on real drives, and it recovers after a fleet change with as few as 20 recorded failures. The same guarantee also tells operators the true price of safety, which on SMART data is high. Future work: per-group (Mondrian) calibration for drive models, online recalibration that detects fleet change from unlabelled data, and sensors richer than SMART.

Reproducibility. All code, the frozen `prereg.json` (hypotheses, settings and SHA-256 hashes of code and data), raw result files and figures are in the project repository; `git tag freeze` marks the commit at which the analysis was frozen. Running the three commands listed in `prereg.json` reproduces every number in this paper.

Use of AI tools. [Student to complete: describe how AI assistants were used, for example in coding, data processing and editing, and confirm that the student designed, ran, checked and can explain every part of the project.]

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